

Titanic Survival Prediction Project Report

This report summarizes the complete workflow implemented in the Titanic notebook, including data exploration, preprocessing, feature engineering, model training, evaluation, and final conclusions.

1. Project Objective

The goal of this project was to analyze the Titanic dataset and build a machine learning model capable of predicting whether a passenger survived or not based on passenger information.

2. Dataset Overview

The dataset contains 891 rows and 12 columns. Important features included passenger class, gender, age, fare, embarkation port, family information, and survival status.

Feature	Description
Pclass	Passenger class
Sex	Passenger gender
Age	Passenger age
Fare	Ticket fare
Embarked	Embarkation port
SibSp / Parch	Family members aboard
Survived	Target variable

3. Data Exploration & Cleaning

- Checked dataset shape, data types, and statistical summary.
- Identified missing values in Age, Fare, and Embarked columns.
- Filled missing numerical values using median values.
- Filled missing categorical values using the mode.
- Verified dataset consistency after preprocessing.

4. Feature Engineering

- Extracted passenger titles from names (Mr, Mrs, Miss, etc.).
- Created FamilySize feature using SibSp and Parch.
- Created IsAlone feature to identify passengers traveling alone.
- Created AgeBand and FareBand categories to simplify continuous variables.
- Dropped unnecessary columns such as PassengerId, Name, Ticket, and Cabin.

5. Encoding & Scaling

Categorical features were encoded using LabelEncoder. Feature scaling was applied using StandardScaler to normalize numerical features before training the machine learning model.

6. Model Building

The XGBoost classifier was selected because of its strong performance in classification tasks. The workflow included:

- Train/Test split using stratified sampling.
- Baseline model evaluation using cross-validation.
- Hyperparameter tuning using GridSearchCV.
- Training the best optimized XGBoost model.

7. Model Performance

Metric	Result
Baseline CV Accuracy	81.89%
Best CV Accuracy	83.43%
Test Accuracy	83.24%
Cross Validation Folds	5 Folds

The optimized model achieved strong and stable performance with good balance between precision and recall.

8. Key Insights from the Data

- Female passengers had a much higher survival rate than males.
- Passengers in higher classes (1st class) had better survival chances.
- Passengers traveling alone were less likely to survive.
- Age and fare both influenced survival probability.
- Feature engineering significantly improved model performance.

9. Model Saving & Deployment Readiness

The final trained model, scaler, encoders, and feature list were saved using joblib. The notebook also verified successful model loading and prediction consistency.

10. Final Conclusion

This project successfully demonstrated a complete machine learning workflow, starting from raw data preprocessing and ending with a well-optimized predictive model. The notebook followed good ML practices including feature engineering, scaling, cross-validation, hyperparameter tuning, and model persistence.